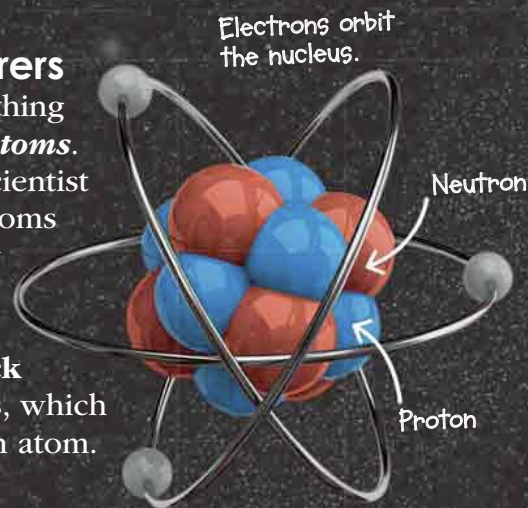


Future nuclear-powered rockets could one day take humans to Mars.

Atomic explorers

By 1900, scientists knew that everything is made from *tiny particles called atoms*.

In 1909, New Zealand-born scientist **Ernest Rutherford** showed that atoms have a central nucleus, orbited by smaller particles called electrons. Later, discoveries by Rutherford and English physicist **James Chadwick** identified protons and neutrons, which make up the nucleus of an atom.



Lise Meitner and Otto Hahn in their German laboratory in 1913.



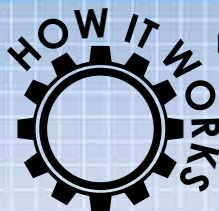
Nuclear fission

The **POWER OF ATOMS** was unleashed in 1938 by scientists **Lise Meitner**, **Otto Hahn**, and **Fritz Strassman**: They split the nucleus of a uranium atom by firing neutrons at it. As the nucleus split, some of its mass was converted into heat, a process that became known as nuclear fission. **Enrico Fermi**, an Italian scientist living in the United States, headed the team that created the first controlled **FISSION CHAIN REACTION** in the world's first **nuclear reactor**.

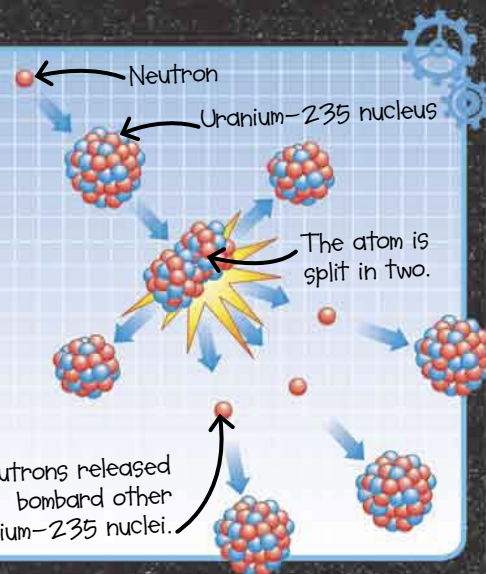
How it changed

Nuclear power already drives some power plants, submarines, and ships. In the future, nuclear-powered spacecraft could carry human beings farther into space than ever before.

the world.



Nuclear fission works by splitting atoms. Some nuclei of uranium-235 naturally split in two, releasing neutrons. Some of the neutrons hit other uranium-235 nuclei, causing them to split, releasing energy and more neutrons, which in turn hit more uranium atoms. This self-sustaining process is called a chain reaction.



USS NAUTILUS, the first **NUCLEAR-POWERED SUBMARINE**, was launched in 1954. The icebreaker **LENIN** was the first **NUCLEAR SURFACE SHIP**.

